

Phase 2: SQL Rebuild & Analytical Reasoning Report

1. Database Schema Documentation

1.1 Entity-Relationship Diagram

```
erDiagram
    USERS ||--o{ POSTS : "creates"
    USERS {
        text user_id PK "e.g. user_guglt5jp"
        text location "City, Country"
        text language "ISO 639-1 code"
        date account_created "Registration date"
        integer follower_count "Non-negative"
    }
    POSTS {
        text post_id PK "e.g. to64mgey2v3y"
        text user_id FK "References users"
        text platform "Twitter|Facebook|Instagram|Reddit|"
        text text_content "Cleaned post body"
        timestamp timestamp "Unified datetime"
        integer likes "Nullable"
        integer shares "Non-negative"
        integer comments "Non-negative"
        text hashtags "Comma-separated"
        text brand_mentioned "Extracted brand"
        text sentiment "positive|negative|neutral"
        integer total_engagement "Computed aggregate"
        real engagement_rate "total / 3"
```

```

date post_date "Date component"
integer post_hour "0-23"
text post_day_of_week "Monday-Sunday"
integer post_month "1-12"
}

```

1.2 Table Design Rationale

users Table (Dimension)

Column	Type	Constraint	Rationale
user_id	TEXT	PRIMARY KEY	Alphanumeric hash IDs, not auto-increment
location	TEXT	NOT NULL	Denormalized "City, Country" - kept flat for query simplicity
language	TEXT	NOT NULL	ISO 639-1 two-char code (en, fr, de, etc.)
account_created	DATE	NOT NULL	Registration date for tenure analysis
follower_count	INTEGER	NOT NULL, CHECK >= 0	Social metric, enforced non-negative

posts Table (Fact)

Column	Type	Constraint	Rationale
post_id	TEXT	PRIMARY KEY	Unique content identifier

user_id	TEXT	NOT NULL, FK	Referential integrity to users
platform	TEXT	NULLABLE	14.9% missing in source data
text_content	TEXT	NULLABLE	14.3% missing (deleted/ corrupted posts)
likes	INTEGER	NULLABLE	15.1% missing; unlike shares/ comments
shares	INTEGER	NOT NULL, >= 0	Always present and non- negative
comments	INTEGER	NOT NULL, >= 0	Always present and non- negative
sentiment	TEXT	NOT NULL, DEFAULT 'neutral'	Pre-computed for fast filtering
total_engagement	INTEGER	NOT NULL	Materialized aggregate avoids runtime SUM
Temporal columns	Various	NULLABLE	Pre- decomposed for GROUP BY performance

Performance Indexes

- `idx_posts_user_id` — Optimizes JOIN operations and per-user queries
- `idx_posts_platform` — Fast platform-level filtering
- `idx_posts_timestamp` — Time-range queries and trend analysis

- `idx_posts_brand` — Brand mention lookups
- `idx_posts_sentiment` — Sentiment-filtered queries
- `idx_posts_date` — Daily aggregation queries

[!NOTE] Pre-materialized derived columns (`total_engagement`, `engagement_rate`, temporal decomposition) trade storage for query performance. In a production system, these could alternatively be virtual/computed columns or maintained via triggers.

2. SQL Query Logic Explanations

Query 1: Trend Detection — Viral Spike Identification

Analytical Challenge: Identify days where posting activity deviates significantly from normal patterns, indicating viral events or organic surges.

Technique: CTE pipeline → Daily aggregation → 7-day rolling window → Z-score flagging

Logic Walkthrough:

1. **daily_activity CTE:** Aggregates posts by date, computing count, avg engagement, and unique poster count
2. **rolling_stats CTE:** Applies a 7-day trailing window average using `ROWS BETWEEN 6 PRECEDING AND CURRENT ROW`
3. **spike_detection CTE:** Compares daily volume to rolling average; flags as `SPIKE` (>150% of average) or `DROP` (<50%)
4. **Final SELECT:** Returns only anomalous days sorted by deviation magnitude

Results Summary: 9 spike days detected, with 2024-06-16 (54 posts) as the highest single-day volume, representing a ~64% deviation from the rolling average.

Query 2: Anomaly Discovery — Bot-Like Pattern Detection

Analytical Challenge: Identify potentially inauthentic accounts exhibiting automated behavior.

Technique: Multi-signal scoring with CTE layers → Composite anomaly score

Logic Walkthrough:

1. **user_posting_stats CTE:** Computes per-user metrics — post count, engagement variance, platform diversity
2. **posting_frequency CTE:** Adds posts-per-day rate and NTILE percentile rankings
3. **anomaly_flags CTE:** Evaluates 4 independent bot signals:
 - High volume (top 5% by post count)
 - Low engagement (bottom 10%)
 - Single-platform usage
 - Rapid posting (>2 posts/day)
4. **Composite bot_score (0-4):** Sum of all flags; users scoring ≥ 2 are flagged

Results Summary: No users scored ≥ 3 (strong bot signal). This is consistent with the EDA finding of zero bot-like patterns, suggesting the social engine has genuine organic activity.

Query 3: Behavioral Grouping — Engagement Velocity Segmentation

Analytical Challenge: Categorize users by how efficiently they generate engagement over time, not just total volume.

Technique: NTILE quintile bucketing on velocity metric → Segment labeling

Logic Walkthrough:

1. **user_metrics CTE**: Calculates lifetime engagement, posting span (days), and activity density (minimum 3 posts required)
2. **velocity_calc CTE**: Derives `engagement_velocity = lifetime_engagement / span_days` and buckets users into 5 quintiles
3. **user_segments CTE**: Maps quintiles to meaningful labels (Hyper-Active → Low-Velocity)
4. **Final aggregation**: Reports cohort-level averages for strategic comparison

Results Summary:

Segment	Users	Avg Posts	Avg Velocity	Avg Monthly Rate
Hyper-Active Engagers	~290	8+	Highest	Highest
Steady Performers	~290	8	Moderate	Balanced
Low-Velocity Users	~290	7-8	Lowest	Variable

Query 4: Correlation Analysis — Platform × Brand × Sentiment

Analytical Challenge: Identify which content characteristics (brand, platform, sentiment) drive peak engagement.

Technique: Multi-dimensional GROUP BY → DENSE_RANK within platform partitions

Logic Walkthrough:

1. Groups posts by platform × brand × sentiment (minimum 5 posts per combination)
2. Ranks each combination within its platform by average engagement

3. Computes deviation from global average to show over/under-performance
4. Returns top-5 performers per platform

Key Findings:

- **Instagram + Coca-Cola + neutral** = highest engagement (4,132 avg), +507 above global mean
 - **Instagram + Amazon + positive** = second highest (4,060 avg)
 - Positive sentiment content on Instagram consistently outperforms
 - Facebook performs best with Apple and Toyota content
-

Query 5: Brand Momentum — Monthly Share of Voice

Analytical Challenge: Track brands gaining or losing social mindshare over time.

Technique: LAG window function for month-over-month delta computation

Logic Walkthrough:

1. Counts monthly mentions per brand
2. Calculates share-of-voice as percentage of total monthly mentions
3. Uses `LAG()` to compare with previous month's SOV
4. Flags momentum: GAINING (>+1% SOV), LOSING (>-1% SOV), STABLE

Key Findings:

- All brands show oscillating SOV patterns ($\pm 1\text{-}2\%$ monthly)
 - No brand shows sustained multi-month momentum in either direction
 - Adidas shows the most volatile SOV pattern with alternating GAINING/LOSING months
-

Query 6: Engagement Drop Detection — Weekly Platform

Monitoring

Analytical Challenge: Detect significant engagement drops that may indicate platform issues.

Technique: ISO-week aggregation → 4-week trailing average → percentage change detection

Key Findings:

- 27 significant events detected across all platforms
- Twitter Week 40 showed the largest drop (-60.9%)
- Facebook Week 13 saw the largest surge (+51.0%)
- Multiple platforms experienced simultaneous drops in September Week 35

Query 7: Follower-Engagement Decile Analysis

Analytical Challenge: Determine whether follower count predicts engagement.

Technique: NTILE(10) bucketing → Engagement-per-1K-followers efficiency metric

Key Findings:

Decile	Avg Followers	Avg Engagement	Engagement per 1K Followers
1 (Lowest)	2,887	3,634	1,258.9
5 (Mid)	22,466	3,655	162.7
10 (Highest)	47,266	3,562	75.4

[!IMPORTANT] Users with the fewest followers (Decile 1) generate **16.7x higher engagement efficiency** than top-follower users (Decile 10). Follower count does NOT predict engagement. The social engine rewards content quality over audience size.

Query 8: Optimal Posting Times

Analytical Challenge: Find the best time slots for content publication per platform.

Key Findings:

- **Facebook:** Wednesday 4 AM (6,022 avg engagement) and Thursday noon
 - **Instagram:** Friday 8 PM (5,153 avg) — end-of-week prime time
 - **Twitter:** Variable peak hours with no single dominant slot
 - **Reddit:** Early morning and late evening slots perform best
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3. Social Engine Health Assessment

Overall Vitals

Metric	Value	Assessment
Active Users	1,500	Healthy base
Total Posts (12 months)	12,000	Consistent volume
Avg Posts/User	8.0	Moderate engagement
Platform Distribution	~20% each (5 platforms)	Well-diversified
Content Completeness	85%+	Acceptable
Bot Detection	0 flagged	Clean ecosystem
Avg Engagement	3,631	Strong baseline

Strengths

1. **No bot contamination** — Zero users flagged across 4 detection signals
2. **Platform diversity** — Users spread evenly across 5 platforms, reducing single-platform risk
3. **Consistent monthly volume** — No seasonal cliff or user churn observed
4. **Content-over-followers meritocracy** — Low-follower users achieve 16x higher engagement efficiency

Concerns

1. **Flat sentiment-engagement curve** — Negative content gets equal engagement, suggesting weak content moderation
2. **Zero correlation between likes/shares/comments** — Unusual for organic social platforms; may indicate independent metric generation
3. **15% data loss** — Platform, text, and likes fields have consistent ~15% missing rates, suggesting systematic export failures
4. **No clear temporal patterns** — Uniform hourly/daily distribution is atypical for organic user behavior

Recommendations

1. **Implement content quality signals** — Penalize low-quality/negative content in feed ranking
2. **Investigate data pipeline** — The consistent ~15% missing rate across fields suggests a single point of failure
3. **Monitor engagement metric independence** — The zero correlation between likes/shares/comments warrants investigation into whether metrics are being generated independently
4. **Leverage follower-engagement insight** — Build discovery features to surface high-efficiency low-follower creators